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Analytical Dynamics of a Monkeypox Model Using Caputo-Fabrizio Fractional Derivative and Sumudu Transform Method

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⁷ When a norm is used in the concept of majorizing, the distinction between sub-sequent terms that implies

⁸ In the following section, we will attempt to establish that the outcomes that were mentioned in the section that came before this one are completely independent. We presume that more sets of results are available for the configuration specified in Eqn. (29).

Abstract: This study presents a fractional-order mathematical model for the transmission dynamics of monkeypox, formulated using the Caputo-Fabrizio fractional derivative to better capture the memory and hereditary properties inherent in real-world biological systems. The model incorporates key epidemiological compartments to reflect the stages of monkeypox infection and transmission. To obtain analytical solutions, the Sumudu transform method is employed due to its efficiency in handling fractional differential equations. The existence and uniqueness of solutions are rigorously verified using fixed-point theory and the Picard–Lindelöf approach, ensuring mathematical consistency and reliability of the model. The proposed framework provides deeper insights into the progression of monkeypox and offers a robust analytical approach for studying disease dynamics under fractional-order settings. This work contributes significantly to understanding the impact of memory effects in disease modelling and can inform future control strategies for emerging infectious diseases like monkeypox.
